Course Code: ITA0608

Course Name : Machine Learning

Experiment 4: Artificial Neural Network using Backpropagation Algorithm

Aim:

To build an Artificial Neural Network (ANN) using the Backpropagation algorithm and test its performance on a sample dataset for classification.

Algorithm:

1. Import the required Python libraries.
2. Load the dataset.
3. Split the dataset into training and testing sets.
4. Normalize the input features.
5. Create an Artificial Neural Network (ANN).
6. Add input, hidden, and output layers.
7. Compile the model using the Adam optimizer and binary cross-entropy loss function.
8. Train the ANN using the Backpropagation algorithm.
9. Test the model using the test dataset.
10. Predict the output and calculate the accuracy.

CODE:

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

# Load Iris Dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42)

# Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create ANN Model
model = MLPClassifier(hidden_layer_sizes=(10,),
                      activation='relu',
                      solver='adam',
                      max_iter=1000,
                      random_state=42)
```

OUTPUT:

```
Actual Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Predicted Values:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Accuracy: 100.0

Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```

Result:

Artificial Neural Network (ANN) was successfully implemented using the Backpropagation algorithm. The model was trained and tested using the given dataset and achieved high classification accuracy, demonstrating the effectiveness of the Backpropagation learning algorithm.